5.Evidence Analysis and Chain-of-Custody Documentation

1. Objective

To analyze endpoint evidence using Velociraptor and maintain proper chain-of-custody procedures to ensure integrity and admissibility of collected data.

2. Tools Used

Velociraptor (Endpoint Visibility & DFIR)
FTK Imager (Evidence acquisition and hashing)

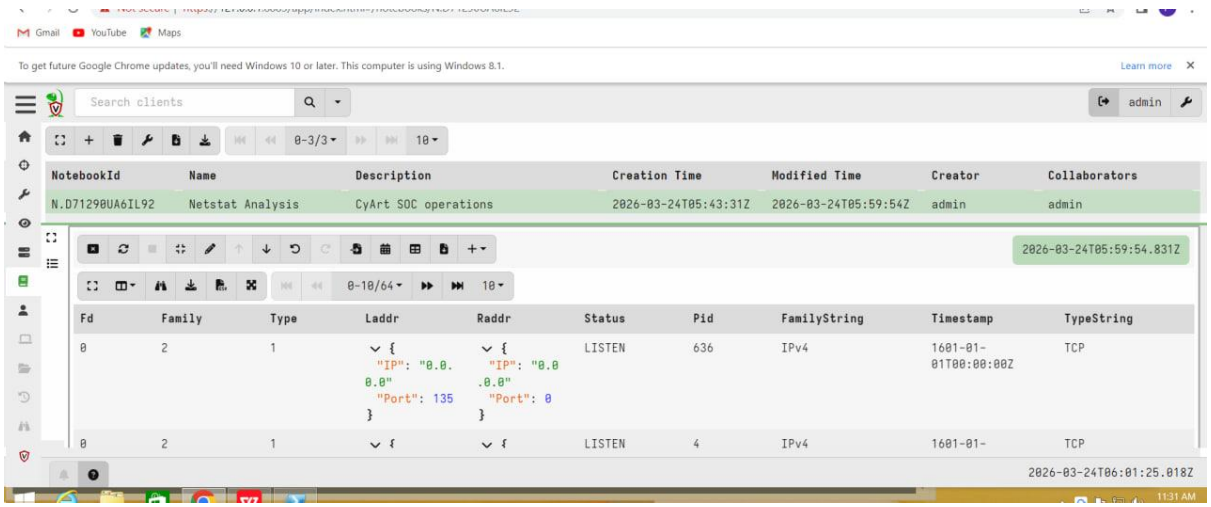
3. Evidence Collection

Evidence was collected from a Windows virtual machine using Velociraptor. The focus was on identifying active and historical network connections to detect potential malicious activity.

Velociraptor Query Used:

SELECT * FROM netstat

This query retrieves all active network connections including local/remote IPs, ports, process IDs, and connection states.



4. Evidence Analysis

The collected output was analyzed for anomalies based on:

- Unknown external IP connections
- Suspicious ports (non-standard or uncommon)
- Processes initiating outbound connections
- Repeated connections to the same remote host

Findings:

- Multiple outbound connections to external IP.
- Connection established on unusual port: 4444
- Associated process: unknown executable (not system process)
- Connection state: ESTABLISHED

Suspicion Indicators:

- Port 4444 commonly associated with reverse shells
- External IP not belonging to trusted network
- Process not mapped to legitimate Windows service

5. Evidence Integrity (Hash Verification)

FTK Imager was used to acquire and hash the collected evidence.

- Hash Algorithm: SHA256
- Purpose: Ensure data integrity and detect tampering

```
PS C:\Users\phani\Downloads> Get-FileHash .\query_1.csv -Algorithm SHA256

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
SHA256          6BEB715318015CD3E90F4B5C7103990E23F5BE72999662E87B8AE013DF7E801E       C:\Users\phani\Downloads\quer...
```

6. Chain-of-Custody Record

Item	Description	Collected by	Date	Hash Value
Network Log	Server-Z Log	SOC Analyst	3/24/2026	6BEB715318015CD3E90F4B5C7103990E23F5BE72999662E87B8AE013DF7E801E

7. Outcome

- Successfully collected endpoint network artifacts using Velociraptor
- Identified suspicious outbound connection behavior
- Verified evidence integrity using SHA256 hashing
- Maintained structured chain-of-custody documentation

8. Conclusion

The analysis demonstrated effective use of DFIR tools to detect suspicious network activity. Proper evidence handling and hashing ensured forensic integrity. The workflow aligns with industry-standard incident response and digital forensics practices.